

Roll Number: _____

Thapar Institute of Engineering and Technology, Patiala

Department of Computer Science and Engineering

MCA (1st Year)

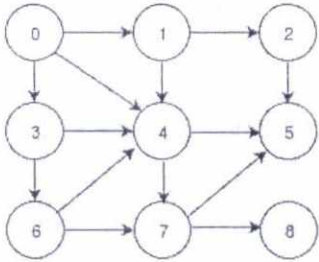
MCA-105: Data Structures

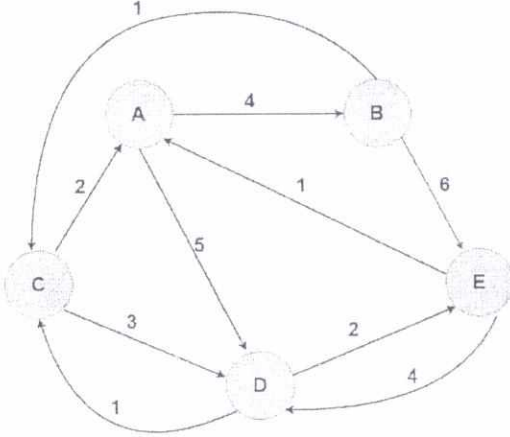
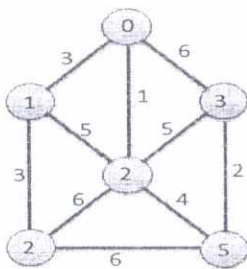
End-Sem Test; Dec 05, 2024

Time: 03Hours; MM: 45

Name of Faculty: Dr. Rinkle Rani

Note : Attempt any five questions.

Q No.	Question	Marks	CO	BL
Q1. a)	Sort the following sequence of elements in descending order by applying Heap Sort. 38 24 44 21 55 47 67 49 For each iteration : i) Design the min heap. ii) Delete the root and redesign the min heap. iii) Show the contents of the array.	(6)	CO2	L3
b)	What is hashing? Explain various hash functions used for hashing with the help of examples.	(3)	CO2	L2
Q2. a)	Find all the nodes approachable from node 0 of the following graph using BFS graph traversal algorithm. 	(6)	CO4	L3
b)	Discuss logic of Radix Sort with example.	(3)	CO2	L2
Q3. a)	Write algorithm for counting sort. Apply your logic on the following array. Show status of count array and output array during each iteration. 2 5 3 0 2 3 0 3	(6)	CO2	L3
b)	Differentiate between the following in a tabular format. i) Array and Link List ii) Stack and Queue	(3)	CO1	L2

Q4. a)	Insert the following keys one by one into a binary search tree in the order specified. 18, 35, 23, 12, 6, 28, 15, 4 i) Show the final binary search tree after the insertions. ii) How much memory is required to store this binary tree sequentially? iii) Write pre-order and post-order traversals of the above BST. iv) Draw the binary search tree after deleting 18 from it.	(6)	CO3	L3
b)	Define Graph. Discuss the various memory representations techniques used for graph with the help of an example.	(3)	CO3	L1
Q5. a)	Apply All Pairs Shortest Path (APSP) Floyd's Algorithm on the following graph. 	(6)	CO4	L3
b)	What do you mean by a doubly link list? Discuss advantages and disadvantages of doubly linked list over singly linked list.	(3)	CO1	L2
Q6. a)	Find out the minimum spanning tree using Prim's Algorithm of the following graph. What is the total weight of the resulting minimum spanning tree? 	(6)	CO3	L3
b)	Define binary tree? Discuss the various memory representations techniques used for binary trees with the help of suitable example.	(3)	CO3	L1